

Zihao Zhu

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📄 Google Scholar (1000+ Citations)

Education

The Chinese University of Hong Kong, Shenzhen <i>Ph.D. in Data Science, School of Data Science</i> Advisor: Baoyuan Wu	Shenzhen, China 2021–2026
University of Chinese Academy of Sciences <i>M.S. in Cybersecurity, Institute of Information Engineering</i> Advisor: Jing Yu	Beijing, China 2018–2021
China University of Mining and Technology <i>B.S. in Computer Science and Technology, School of Computer Science and Technology</i>	Xuzhou, China 2014–2018

Research Interests

Large Language Model Safety, Reasoning Models, Embodied AI Agents, Data Governance, AI Security & Privacy, Multimodal Learning

Research Experience

University at Buffalo, SUNY <i>Postdoctoral Researcher</i>	Buffalo, NY, USA 2026–Present
○ Continuing research on trustworthy AI, large language models, and AI safety.	
Tencent AI Lab <i>Research Intern</i>	Shenzhen, China Mar. 2024–July 2024
○ Led the design of EARBench, the first physical risk awareness evaluation benchmark for embodied AI. Built EARDataset using multi-agent collaboration to generate risk scenarios for task planning safety evaluation.	

Publications

Published/accepted 18 journal, conference, and workshop papers (6 as first author); 4 preprints (3 as first author)

(TPAMI: 3; IJCV: 2; NeurIPS: 3; ICLR: 3; CVPR: 1; IJCAI: 2; AAAI: 1; ICASSP: 2; PR: 1)

Large Language Model Safety & Chain-of-Thought

- To Think or Not to Think: Exploring the Unthinking Vulnerability in Large Reasoning Models
Zihao Zhu, Hongbao Zhang, Ruotong Wang, Xu Ke, Siwei Lyu, Baoyuan Wu (NeurIPS 2025 Workshop)
- Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers
Zihao Zhu contributed as one of the project participants (NeurIPS 2025 Workshop)
- AdvChain: Adversarial Chain-of-Thought Tuning for Robust Safety Alignment of Large Reasoning Models
Zihao Zhu, Xinyu Wu, Gehan Hu, Siwei Lyu, Ke Xu, Baoyuan Wu (ICLR 2026)

AI Agents

4. EARBench: Towards Evaluating Physical Risk Awareness for Task Planning of Foundation Model-based Embodied AI Agents
Zihao Zhu, Bingzhe Wu, Zhengyou Zhang, Lei Han, Qingshan Liu, Baoyuan Wu ([arXiv 2024](#))
5. BrandFusion: A Multi-Agent Framework for Seamless Brand Integration in Text-to-Video Generation
Zihao Zhu, Ruotong Wang, Siwei Lyu, Min Zhang, Baoyuan Wu ([CVPR 2026 Findings](#))

Data Governance

6. VDC: Versatile Data Cleanser based on Visual-Linguistic Inconsistency by Multimodal Large Language Models
Zihao Zhu, Mingda Zhang, Shaokui Wei, Bingzhe Wu, Baoyuan Wu ([ICLR 2024](#))
7. HMGIE: Hierarchical and Multi-Grained Inconsistency Evaluation for Vision-Language Data Cleansing
Zihao Zhu, Hongbao Zhang, Guanzong Wu, Siwei Lyu, Baoyuan Wu ([arXiv 2024](#))

AI Security

8. Attacks in Adversarial Machine Learning: A Systematic Survey from the Life-cycle Perspective
Baoyuan Wu, **Zihao Zhu**, Li Liu, Qingshan Liu, Zhaofeng He, Siwei Lyu ([IJCV 2025](#))
9. Defenses in Adversarial Machine Learning: A Survey
Baoyuan Wu, Mingli Zhu, Meixi Zheng, **Zihao Zhu**, Shaokui Wei, Mingda Zhang, Hongrui Chen, Danni Yuan, Li Liu, Qingshan Liu ([IEEE TPAMI 2025](#))
10. BlackboxBench: A Comprehensive Benchmark of Black-box Adversarial Attacks
Meixi Zheng, Xuanchen Yan, **Zihao Zhu**, Hongrui Chen, Baoyuan Wu ([IEEE TPAMI 2025](#))
11. BackdoorBench: A Comprehensive Benchmark and Analysis of Backdoor Learning
Baoyuan Wu, Hongrui Chen, Mingda Zhang, **Zihao Zhu**, Shaokui Wei, Danni Yuan, Mingli Zhu, Ruotong Wang, Li Liu, Chao Shen ([IJCV 2025](#))
12. Versatile Backdoor Attack with Visible, Semantic, Sample-Specific, and Compatible Triggers
Ruotong Wang, Hongrui Chen, **Zihao Zhu**, Li Liu, Baoyuan Wu ([IEEE TPAMI 2024](#))
13. Reliable Poisoned Sample Detection against Backdoor Attacks Enhanced by Sharpness Aware Minimization
Mingda Zhang, Mingli Zhu, **Zihao Zhu**, Baoyuan Wu ([ICLR 2026](#))
14. BackdoorBench: A Comprehensive Benchmark of Backdoor Learning
Baoyuan Wu, Hongrui Chen, Mingda Zhang, **Zihao Zhu**, Shaokui Wei, Danni Yuan, Hongyuan Zha ([NeurIPS 2022](#))
15. Boosting Backdoor Attack with a Learnable Poisoning Sample Selection Strategy
Zihao Zhu, Mingda Zhang, Shaokui Wei, Li Shen, Yanbo Fan, Baoyuan Wu ([arXiv 2023](#))

Multimodal Learning

16. From Shallow to Deep: Compositional Reasoning over Graphs for Visual Question Answering
Zihao Zhu ([ICASSP 2022](#))
17. Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering
Zihao Zhu, Jing Yu, Yujing Wang, Yajing Sun, Yue Hu, Qi Wu ([IJCAI 2020](#), [177 citations](#))
18. Cross-Modal Knowledge Reasoning for Knowledge-based Visual Question Answering

Jing Yu, **Zihao Zhu**, Yujing Wang, Weifeng Zhang, Yue Hu, Jianlong Tan (Pattern Recognition 2020)

Other Publications.....

19. Learning to Optimize Permutation Flow Shop Scheduling via Graph-based Imitation Learning
Longkang Li, Siyuan Liang, **Zihao Zhu**, Xiaochun Cao, Chris Ding, Hongyuan Zha, Baoyuan Wu (AAAI 2024)
20. Enhanced Few-Shot Class-Incremental Learning via Ensemble Models
Mingli Zhu, **Zihao Zhu**, Sihong Chen, Chen Chen, Baoyuan Wu (arXiv 2024)
21. MCR-NET: A Multi-Step Co-Interactive Relation Network for Unanswerable Questions on Machine Reading Comprehension
Wei Peng, Yue Hu, Luxi Xing, Yuqiang Xie, **Zihao Zhu**, Yajing Sun (ICASSP 2021)
22. DAM: Deliberation, Abandon and Memory Networks for Generating Detailed and Non-repetitive Responses
Xiaozhe Jiang, Jing Yu, Zengchang Qin, **Zihao Zhu**, Qi Wu (IJCAI 2020)

Academic Service

Reviewer: IEEE TIP, ICLR, ICML, NeurIPS, CVPR, ICCV, AAAI, AISTATS

Awards & Honors

2024: Guotai Junan Scholarship

2021: AIRS Talent of PhD Research Program

2020: University of Chinese Academy of Sciences Merit Student

2018: Outstanding Undergraduate Thesis Award

2015, 2016: The First Prize Scholarship of CUMT